

NOW THAT WE KNOW THE Z-TRANSFORM FOR SIGNALS, LET'S USE IT TO ANALYZE LTI SYSTEMS

CAUSALITY

FIRSTLY, WE CAN EASILY ASSESS THE CAUSALITY OF A SYSTEM BY LOOKING AT THE ROC OF ITS IMPULSE RESPONSE.

$$H(z), |z| > \alpha \Rightarrow \text{CAUSAL}$$

$$H(z), |z| < \alpha \Rightarrow \text{ANTICAUSAL}$$

$$H(z), \alpha < |z| < \beta \Rightarrow \text{TWO-SIDED}$$

STABILITY

WE HAVE SEEN THAT A SYSTEM IS STABLE, IF ITS IMPULSE RESPONSE IS ABSOLUTELY SUMMABLE. BUT WE ALSO KNOW THAT THE ABSOLUTE SUMMABILITY IS ONE OF THE SUFFICIENT CONDITIONS THAT GUARANTEE CONVERGENCE OF THE DFT, MEANING THE UNITARY CIRCLE WILL BE INCLUDED IN THE ROC.

$$h[n] \text{ STABLE} \Rightarrow \sum_{n=-\infty}^{+\infty} |h[n]| < \infty \Rightarrow \text{DFT CONVERGES} \Rightarrow r=1 \in \text{ROC}$$

AND THEREFORE

$$\text{UNIT CIRCLE} \in \text{ROC} \Rightarrow \text{STABILITY}$$

STABILITY AND CAUSALITY

IF WE WANT CAUSALITY AND STABILITY, THE OUTER-MOST POLE MUST LIE WITHIN THE UNIT CIRCLE, AND ALL THE POLES MUST LIE INSIDE THE UNIT CIRCLE

$$|z| > \alpha, \alpha < 1$$

RECURSIVE/LCCDE SYSTEMS

THE Z-TRANSFORM CAN BE USED WHEN DEALING WITH LCCDES TO SIMPLIFY SHIT.

$$\sum_{k=0}^N a_k Y[n-k] = \sum_{k=0}^M b_k x[n-k]$$

INSTEAD OF APPLYING THE COMPLEX METHODS SEEN BEFORE, WE CAN SIMPLY APPLY THE Z-TRANSFORM

$$\sum_{k=0}^N a_k Y(z) z^{-k} = \sum_{k=0}^M b_k X(z) z^{-k} \rightarrow Y(z) \sum_{k=0}^N a_k z^{-k} = X(z) \sum_{k=0}^M b_k z^{-k} \rightarrow \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^M b_k z^{-k}}{\sum_{k=0}^N a_k z^{-k}} = H(z)$$

JUST THE COEFFICIENTS!

EX

$$Y[n] = \alpha Y[n-1] + b x[n], |\alpha| < 1, \text{causal}$$

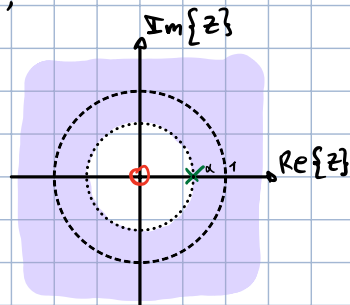
↓

$$Y(z) = \alpha Y(z) z^{-1} + b X(z) \rightarrow Y(z) - \alpha Y(z) z^{-1} = b X(z) \rightarrow Y(z) (1 - \alpha z^{-1}) = b X(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b}{1 - \alpha z^{-1}} = b \frac{z}{z - \alpha}, \quad |z| > \alpha, \quad \alpha < 1$$

DUE TO INITIAL CAUSALITY REQUIREMENT

$$h[n] = b a^n u[n], \quad a < 1$$



EX

$$H(z) = \frac{(1+z^{-1})^2}{(1-\frac{1}{2}z^{-1})(1+\frac{3}{4}z^{-1})} \quad Y[n] = ? \quad , \text{ causal}$$

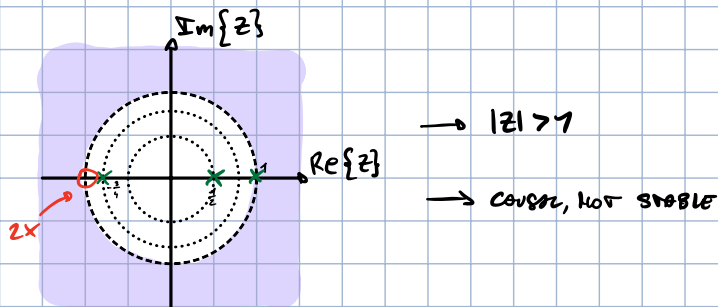
$$H(z) = \frac{1+z^{-2}+2z^{-1}}{1+\frac{3}{4}z^{-1}-\frac{1}{2}z^{-1}-\frac{3}{8}z^{-2}} = \frac{1+2z^{-1}+z^{-2}}{1+\frac{1}{4}z^{-1}-\frac{3}{8}z^{-2}} = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^M b_k z^{-k}}{\sum_{k=0}^N d_k z^{-k}}$$

$$Y[n] + \frac{1}{4}Y[n-1] - \frac{3}{8}Y[n-2] = x[n] + 2x[n-1] + x[n-2]$$

SUPP. $x[n] = u[n], \quad Y[n < 0] = 0$

$$Y(z) + \frac{1}{4}Y(z)z^{-1} - \frac{3}{8}Y(z)z^{-2} = X(z) + 2X(z)z^{-1} + X(z)z^{-2}$$

$$Y(z) = H(z)X(z) = \frac{(1+z^{-1})^2}{(1-\frac{1}{2}z^{-1})(1+\frac{3}{4}z^{-1})} \quad X(z) = \frac{(1+z^{-1})^2}{(1-\frac{1}{2}z^{-1})(1+\frac{3}{4}z^{-1})} \frac{1}{1-z^{-1}} = \frac{(1+z^{-1})^2}{(1-\frac{1}{2}z^{-1})(1+\frac{3}{4}z^{-1})(1-z^{-1})}$$



WE CAN ALSO OBTAIN THE INVERSE TRANSFORM BECAUSE WE CAN DEDUCE WHAT ROC TO USE FOR EACH OF THE THREE COMPONENTS TO REACH THE OVERALL ROC.

$$|z| > \frac{1}{2} \cap |z| > \frac{3}{4} \cap |z| > 1$$

BY PARTIAL FRACTION EXPANSION

$$\frac{A}{(1-\frac{1}{2}z^{-1})} + \frac{B}{(1+\frac{3}{4}z^{-1})} + \frac{C}{(1-z^{-1})}$$

$$A = Y(z) \cdot \left(1 - \frac{1}{2}z^{-1}\right) \Big|_{z^{-1}=2} = \frac{(1+z^{-1})^2}{\left(1+\frac{3}{4}z^{-1}\right)(1-z^{-1})} \Big|_{z^{-1}=2} = \frac{(1+2)^2}{\left(1+\frac{3}{2}\right)(1-2)} = \frac{9}{\frac{5}{2} \cdot (-1)} = -\frac{18}{5}$$

$$B = Y(z) \cdot \left(1 + \frac{3}{4}z^{-1}\right) \Big|_{z^{-1}=-\frac{4}{3}} = \frac{(1+z^{-1})^2}{\left(1-\frac{1}{2}z^{-1}\right)(1-z^{-1})} \Big|_{z^{-1}=-\frac{4}{3}} = \frac{\left(1-\frac{4}{3}\right)^2}{\left(1+\frac{1}{2}\frac{4}{3}\right)\left(1+\frac{4}{3}\right)} = \frac{\left(-\frac{1}{3}\right)^2}{\frac{5}{3} \cdot \frac{7}{3}} = \frac{\frac{1}{9}}{\frac{35}{9}} = \frac{1}{35}$$

$$C = Y(z) \cdot (1-z^{-1}) \Big|_{z^{-1}=1} = \frac{(1+z^{-1})^2}{\left(1-\frac{1}{2}z^{-1}\right)\left(1+\frac{3}{4}z^{-1}\right)} \Big|_{z^{-1}=1} = \frac{4}{\frac{1}{2} \cdot \frac{7}{4}} = \frac{32}{7}$$

$$Y(z) = \frac{-\frac{18}{5}}{\left(1-\frac{1}{2}z^{-1}\right)} + \frac{\frac{1}{35}}{\left(1+\frac{3}{4}z^{-1}\right)} + \frac{\frac{32}{7}}{(1-z^{-1})}$$

$$= -\frac{18}{5} \frac{z}{z-\frac{1}{2}} + \frac{1}{35} \frac{z}{z+\frac{3}{4}} + \frac{32}{7} \frac{z}{z-1}$$

$$\rightarrow Y[n] = -\frac{18}{5} \left(\frac{1}{2}\right)^n \mu[n] + \frac{1}{35} \left(-\frac{3}{4}\right)^n \mu[n] + \frac{32}{7} \mu[n]$$

STABILITY AND POLES

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^M b_k z^{-k}}{\sum_{k=0}^N d_k z^{-k}} = \sum_{k=0}^{M-N} A_k z^{-k} + \sum_{k=1}^N \frac{B_k}{1 - p_k z^{-1}}$$

IN CASE OF NO MULTIPLICITY
SINGLE POLES

≠ 0 (IF M > N)
RESULT OF LONG DIVISION

POLES

IF $h[n]$ IS CAUSAL, THEN BY APPLYING THE INVERSE TRANSFORM WE HAVE

$$h[n] = \sum_{k=0}^{M-N} A_k \delta[n-k] + \sum_{k=1}^N B_k p_k^n \mu[n]$$

AND IF WE WANT ALSO STABILITY, $h[n]$ MUST BE ABSOLUTELY SUMMABLE $\Rightarrow |p_k| < 1 \Rightarrow$ ALL POLES INSIDE UNIT CIRCLE.

HENCE A CAUSAL LTI SYSTEM WITH A RATIONAL TRANSFER FUNCTION IS STABLE IF AND ONLY IF ALL POLES OF $H(z)$ ARE INSIDE THE UNIT CIRCLE. ZEROS CAN BE ANYWHERE.

EX

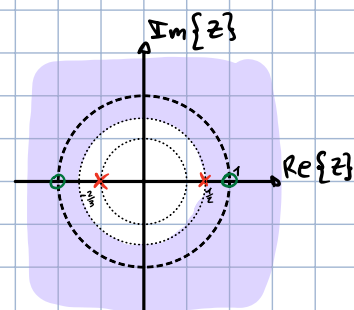
$$H(z) = \frac{1-z^2}{1-\frac{1}{6}z^{-1}-\frac{1}{3}z^{-2}}, \text{ CAUSAL}$$

$$= \frac{z^2-1}{z^2-\frac{1}{6}z-\frac{1}{3}} = \frac{(z-1)(z+1)}{(z+\frac{1}{2})(z-\frac{2}{3})}$$

ZEROS: -1, 1

POLES: $-\frac{1}{2}, \frac{2}{3} \rightarrow < 1 \Rightarrow$ STABLE

IS IT STABLE?



$$\frac{\frac{1}{6} + 1\frac{1}{36} + \frac{1}{3}}{2} < \frac{-\frac{1}{2}}{\frac{2}{3}}$$

$$z^2 - 1 = z^2 - 1^2 = (z+1)(z-1)$$

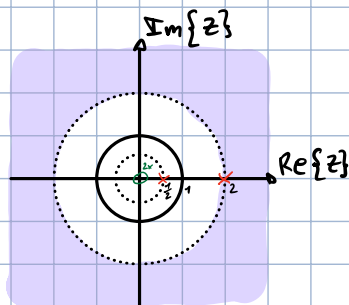
EX

$$y[n] - \frac{5}{2} y[n-1] + y[n-2] = x[n] \quad \text{STABLE?}$$

$$H(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{\sum_{k=0}^N a_k z^{-k}} = \frac{1}{1 - \frac{5}{2} z^{-1} + z^{-2}} = \frac{z^2}{z^2 - \frac{5}{2} z + 1} = \frac{z^2}{(z - \frac{1}{2})(z - 2)}$$

ZEROS: 0, 0

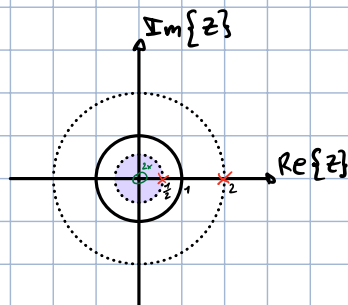
POLES: $\frac{1}{2}, 2$ → CAN'T SAY ANYTHING ABOUT STABILITY YET
SINCE CAUSALITY ISN'T GIVEN.
POLES IN $|z|=1$ IS OK ONLY FOR CAUSAL SYSTEMS
WE CAN SAY THAT CAUSAL VERSION IS SURELY NOT STABLE



CAUSAL

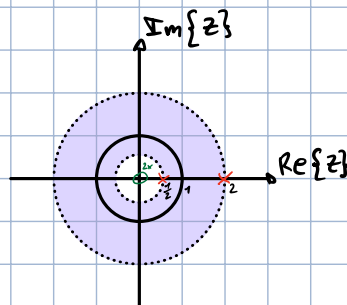
$|z|=1 \notin \text{ROC} \Rightarrow \text{NOT STABLE}$

ALSO HERE WE CAN SAY THAT NOT ALL POLES ARE WITHIN $|z|=1$, SAME THING



ANTICAUSAL

$|z|=1 \notin \text{ROC} \Rightarrow \text{NOT STABLE}$



TWO-SIDED

$|z|=1 \in \text{ROC} \Rightarrow \text{STABLE}$

ONE-SIDED z-TRANSFORM

$$\mathcal{Z}^+ \{x[n]\} := \mathcal{Z} \{x[n] \cdot u[n]\}$$

$$\mathcal{Z} \{x[n]\} = \sum_{n=-\infty}^{+\infty} x[n] z^{-n} = X(z)$$

$$\mathcal{Z}^+ \{x[n]\} = \sum_{n=0}^{\infty} x[n] z^{-n} = X^+(z)$$

OF COURSE, IF THE ORIGINAL SEQUENCE IS CAUSAL, $\mathcal{Z}^+ \{ \} = \mathcal{Z} \{ \}$

MAGIC PROPERTY

$$\begin{aligned} \mathcal{Z}^+ \{x[n-k]\} &= \sum_{n=0}^{\infty} x[n-k] z^{-n} = \sum_{m=-k}^{\infty} x[m] z^{-m-k} = \sum_{m=-k}^{-1} x[m] z^{-m-k} + \sum_{m=0}^{\infty} x[m] z^{-m-k} = \sum_{m=-k}^{-1} x[m] z^{-m-k} + z^{-k} \sum_{m=0}^{\infty} x[m] z^{-m} \\ &= \sum_{m=-k}^{-1} x[m] z^{-m-k} + z^{-k} X^+(z) = \underbrace{x[-k] + x[-k+1] z^{-1} + \dots + x[-1] z^{1-k}}_{\text{INITIAL CONDITIONS}} + z^{-k} X^+(z) \end{aligned}$$

THIS IS USEFUL WHEN SOLVING LCCDES BECAUSE YES

EX

$$y[n] - 2y[n-1] = x[n]$$

$$x[n] = \mu[n]$$

$$y[-1] = 2$$

$$Y^+(z) - \alpha [y[-1] + Y^+(z) z^{-1}] = X^+(z)$$

$$Y^+(z) - \alpha y[-1] - \alpha Y^+(z) z^{-1} = X^+(z)$$

$$Y^+(z) (1 - \alpha z^{-1}) = X^+(z) + \alpha y[-1] = X^+(z) + 2\alpha$$

$$Y^+(z) = \frac{X^+(z)}{1 - \alpha z^{-1}} + \frac{2\alpha}{1 - \alpha z^{-1}} \quad \left\{ \begin{array}{l} Y^+(z) = \frac{1}{(1 - z^{-1})(1 - \alpha z^{-1})} + \frac{2\alpha}{1 - \alpha z^{-1}} = \frac{A}{1 - z^{-1}} + \frac{B}{1 - \alpha z^{-1}} + \frac{2\alpha}{1 - \alpha z^{-1}} \\ X^+(z) = X(z) = \frac{1}{1 - z^{-1}}, |z| > 1 \end{array} \right.$$

$$\left. \begin{array}{l} A = \frac{1}{(1 - \cancel{z^{-1}})(1 - \alpha \cancel{z^{-1}})} \bigg|_{\cancel{z^{-1}} = 1} = \frac{1}{1 - \alpha} \\ B = \frac{1}{(1 - \cancel{z^{-1}})(1 - \alpha \cancel{z^{-1}})} \bigg|_{\cancel{z^{-1}} = \frac{1}{\alpha}} = \frac{1}{1 - \frac{1}{\alpha}} = \frac{\alpha}{\alpha - 1} \end{array} \right\} Y^+(z) = \frac{1}{1 - \alpha} \frac{1}{1 - z^{-1}} + \frac{\alpha}{\alpha - 1} \frac{1}{1 - \alpha z^{-1}} + \frac{2\alpha}{1 - \alpha z^{-1}}$$

$$y[n] \mu[n] = \frac{1}{1 - \alpha} \mu[n] + \frac{\alpha}{\alpha - 1} \alpha^n \mu[n] + 2\alpha \alpha^n \mu[n] = y[n], n \geq 0$$

ex

$$y[n] - 5y[n-1] + 4y[n-2] = 3x[n-1] + 5x[n-2]$$

$$x[n] = 2^n \mu[n]$$

$$y[-1] = 1$$

$$y[-2] = -1$$

$$Y^+(z) - 5[y[-1] + z^{-1}Y^+(z)] + 4[y[-2] + y[-1]z^{-1} + z^{-2}Y^+(z)] = 3[x[-1] + z^{-1}X^+(z)] + 5[x[-2] + x[-1]z^{-1} + z^{-2}X^+(z)]$$

$$Y^+(z) - 5y[-1] + 5z^{-1}Y^+(z) + 4y[-2] + 4z^{-1}y[-1] + 4z^{-2}Y^+(z) = 3\cancel{x[-1]} + 3z^{-1}X^+(z) + 5\cancel{x[-2]} + 5\cancel{x[-1]}z^{-1} + 5z^{-2}X^+(z)$$

$$Y^+(z) - 5 + 5z^{-1}Y^+(z) - 4 + 4z^{-1} + 4z^{-2}Y^+(z) = 3z^{-1}X^+(z) + 5z^{-2}X^+(z)$$

$$4z^{-1} - 9 + (1 - 5z^{-1} + 4z^{-2})Y^+(z) = (3z^{-1} + 5z^{-2})X^+(z)$$

$$Y^+(z) (1 - 5z^{-1} + 4z^{-2}) = X^+(z) (3z^{-1} + 5z^{-2}) - 4z^{-1} + 9$$

$$\left. \begin{array}{l} Y^+(z) = \frac{X^+(z) (3z^{-1} + 5z^{-2}) - 4z^{-1} + 9}{1 - 5z^{-1} + 4z^{-2}} \\ X^+(z) = X(z) = \frac{z}{z - 2} = \frac{1}{1 - 2z^{-1}}, |z| > 2 \end{array} \right\} Y^+(z) = \frac{3z^{-1} + 5z^{-2} + (-4z^{-1} + 9)(1 - 2z^{-1})}{(1 - 2z^{-1})(1 - 5z^{-1} + 4z^{-2})}$$

$$z^2 - 5z + 4 \rightarrow \frac{5 \pm \sqrt{25 - 16}}{2} \left(\begin{array}{l} 1 \\ 4 \end{array} \right)$$

$$Y^+(z) = \frac{3z^{-1} + 5z^{-2} - 4z^{-1} + 8z^{-2} + 9 - 18z^{-1}}{(1 - 2z^{-1})(1 - z^{-1})(1 - 4z^{-1})} = \frac{9 - 19z^{-1} + 13z^{-2}}{(1 - 2z^{-1})(1 - z^{-1})(1 - 4z^{-1})} \rightarrow \frac{A}{1 - 2z^{-1}} + \frac{B}{1 - z^{-1}} + \frac{C}{1 - 4z^{-1}}$$

$$A = \frac{9 - 19z^{-1} + 13z^{-2}}{(1 - \cancel{2z^{-1}})} \bigg|_{\cancel{z^{-1}} = \frac{1}{2}} = \frac{9 - \frac{19}{2} + \frac{13}{4}}{1} = -\frac{11}{4}$$

$$B = \frac{9 - 19z^{-1} + 13z^{-2}}{(1-2z^{-1})(1-z^{-1})(1-4z^{-1})} \bigg|_{z^{-1}=1} = \frac{9 - 19 + 13}{(1-2)(1-4)} = 1$$

$$C = \frac{9 - 19z^{-1} + 13z^{-2}}{(1-2z^{-1})(1-z^{-1})(1-4z^{-1})} \bigg|_{z^{-1}=\frac{1}{4}} = \frac{9 - \frac{19}{4} + \frac{13}{16}}{(1-\frac{1}{2})(1-\frac{1}{4})} = \frac{27}{2}$$

$$Y^*(z) = \frac{-\frac{17}{2}}{1-2z^{-1}} + \frac{1}{1-z^{-1}} + \frac{\frac{27}{2}}{1-4z^{-1}} \rightarrow -\frac{17}{2} 2^n u[n] + u[n] + \frac{27}{2} 4^n u[n], n \geq 0$$

IS THIS SYSTEM STABLE?

$$Y[n] - 5Y[n-1] + 4Y[n-2] = 3x[n-1] + 5x[n-2]$$

$$H(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{\sum_{k=0}^N a_k z^{-k}} = \frac{3z^{-1} + 5z^{-2}}{1 - 5z^{-1} + 4z^{-2}} = \frac{5 + 3z}{z^2 - 5z + 4} = 3 \frac{z + \frac{5}{3}}{(z-4)(z-1)}$$

Zeros: $-\frac{5}{3}, \infty$
Poles: 4, 1

$$\frac{5 \pm \sqrt{25-16}}{2} < 4$$

CAN'T BE STABLE BECAUSE THERE IS A POLE IN 1

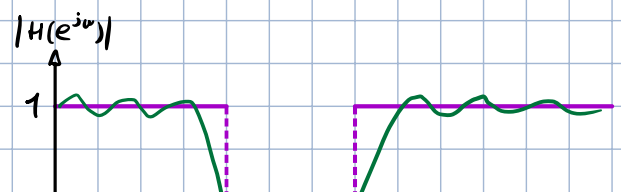
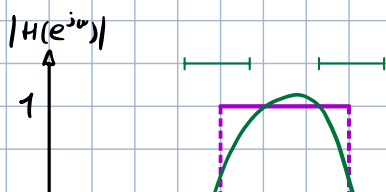
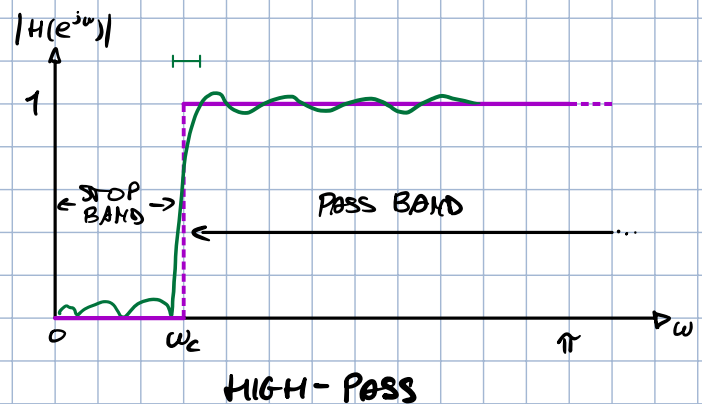
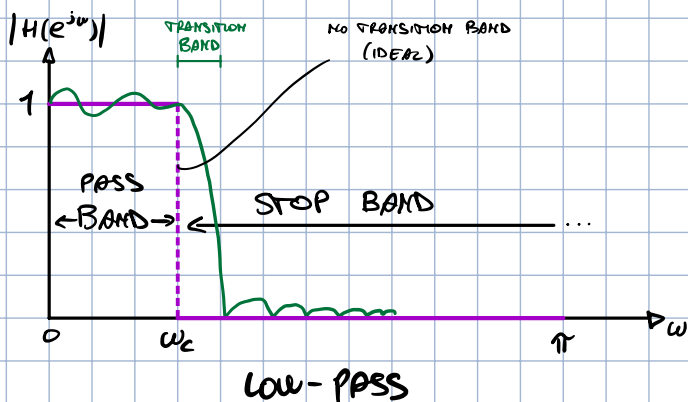
FILTERS

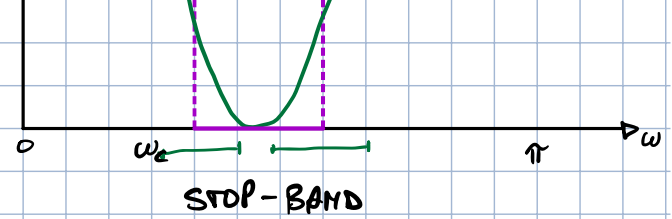
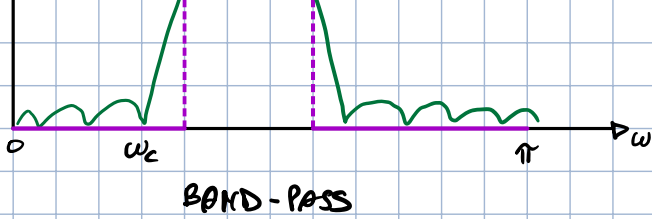
FILTER = IMPULSE RESPONSE OF LTI SYSTEM.

IDEAL VS. REAL FILTERS

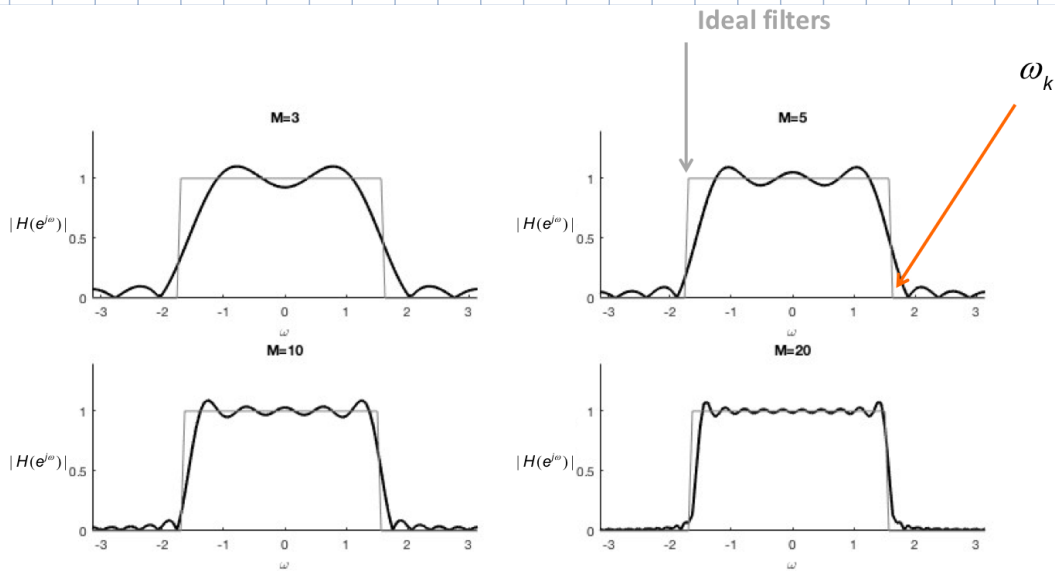
IDEAL FILTERS ARE USEFUL IN THE DESIGN PHASE, BUT IN PRACTICE THEY DON'T EXIST. PRACTICAL FILTERS ARE APPROXIMATIONS.

WE'LL CONSIDER FREQUENCY SELECTIVE FILTERS: PASS SPECIFIC FREQUENCY COMPONENTS WITHOUT SIGNIFICANT DISTORTION WHILE ELIMINATING OTHERS.





IN PRACTICAL FILTERS THE RESPONSE CAN'T BE PERFECTLY FLAT, THE STOP-BAND CAN'T PERFECTLY ELIMINATE FREQUENCY BANDS, TRANSITIONS CAN'T BE INSTANTANEOUS AND THE CUTOFF BAND IS FINITE. DUE TO INSEPARABLE TRANSITIONS
THE IMPULSE RESPONSES OF THESE IDEAL FILTERS ARE NON-CAUSAL AND OF INFINITE LENGTH $\rightarrow$ IMPOSSIBLE



FILTER DESIGN

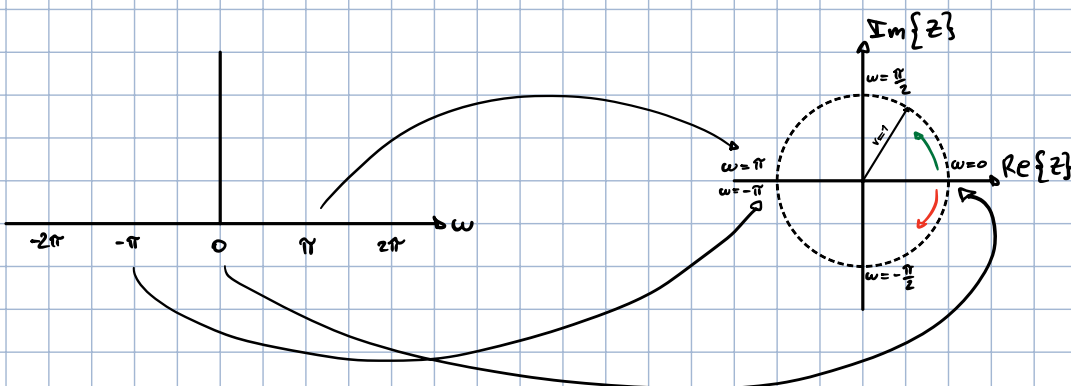
$$\sum_{k=0}^N d_k y[n-k] = \sum_{k=0}^M b_k x[n-k] \rightarrow \sum_{k=0}^N d_k Y(z) z^{-k} = \sum_{k=0}^M b_k X(z) z^{-k} \rightarrow H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^M b_k X(z) z^{-k}}{\sum_{k=0}^N d_k Y(z) z^{-k}}$$

WE ALWAYS WANT A STABLE SYSTEM (AND CAUSAL). $\Rightarrow r=1 \in \text{ROC}$ ALSO, SINCE WE ARE WORKING WITH FREQUENCY FILTERS, IT MAKES SENSE TO WORK IN THE FREQUENCY DOMAIN (DFT).

$$H(e^{j\omega}) = H(z) \Big|_{z=e^{j\omega}} = \frac{\sum_{k=0}^M b_k e^{-j\omega_k}}{\sum_{k=0}^N d_k e^{-j\omega_k}}$$

QUICK INTERLUDE ABOUT PROJECTING THE ω DOMAIN ONTO THE z PLANE

THE PROJECTION OF THE FREQUENCY DOMAIN ONTO THE z PLANE IS SIMPLY THE UNIT CIRCLE ITSELF, BUT WITH A CATCH: INSTEAD OF WORKING IN $[0, 2\pi]$, WE CONSIDER $[-\pi, \pi]$. STARTING FROM 0 WE GO COUNTER CLOCKWISE FROM 0 TO π AND WE GO CLOCKWISE FROM 0 TO $-\pi$.



TO HAVE REAL-COEFFICIENT FILTERS THE AMPLITUDE RESPONSE MUST BE SYMMETRIC W.R.T. $\omega=0$, THEREFORE ON THE Z-PLANE THERE MUST BE SYMMETRY W.R.T. THE REAL AXIS.

CONTINUATION

WE CAN REWRITE THE RATIO TO MAKE IT EASIER TO SHAPE THE RESPONSE BY SIMPLY PLACING POLES AND ZEROS.

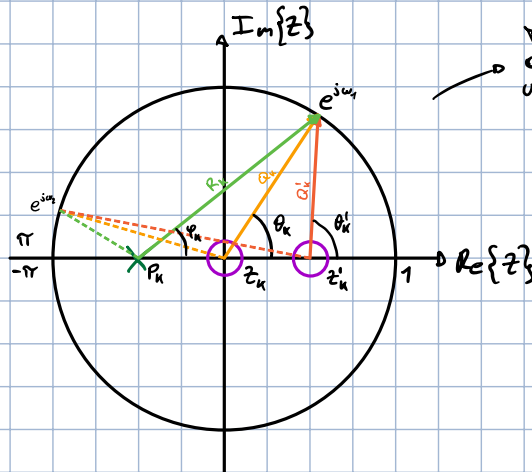
$$H(e^{j\omega}) = \frac{b_0 \prod_{k=1}^M (1 - z_k e^{-j\omega})}{a_0 \prod_{k=1}^N (1 - p_k e^{-j\omega})} = \frac{b_0 \prod_{k=1}^M e^{-j\omega} (e^{j\omega} - z_k)}{a_0 \prod_{k=1}^N e^{-j\omega} (e^{j\omega} - p_k)} = \frac{b_0}{a_0} \frac{e^{-j\omega M}}{e^{-j\omega N}} \frac{\prod_{k=1}^M (e^{j\omega} - z_k)}{\prod_{k=1}^N (e^{j\omega} - p_k)} = \underbrace{\frac{b_0}{a_0} e^{-j\omega(N-M)}}_{\text{AMPLITUDE COEFFICIENT}} \underbrace{\frac{\prod_{k=1}^M (e^{j\omega} - z_k)}{\prod_{k=1}^N (e^{j\omega} - p_k)}}_{\text{PHASE COEFFICIENT}}$$

$$e^{j\omega} - z_k = z - z_k = \rho_k e^{j\theta_k}$$

$$e^{j\omega} - p_k = z - p_k = \rho_k e^{j\theta_k}$$

ρ_k $\leftarrow$ DISTANCE
 θ_k $\leftarrow$ ANGLE/PHASE

IT'S JUST THE DISTANCE BETWEEN THE POLE AND A POINT DEFINED BY ω ON THE UNIT CIRCLE



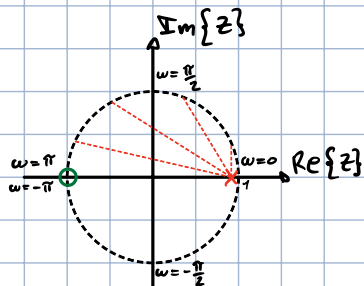
THE PROJECTION OF THE ω DOMAIN ON THE Z PLANE IS SIMPLY THE UNIT CIRCLE ITSELF.

NOTE THAT IF THERE IS A POLE/ZERO IN $z=0$, IT'S DISTANCE FROM ANY POINT IN THE UNIT CIRCLE OBVIOUSLY NEVER CHANGES.

LITTLE CONCEPT EXPLANATION TO UNDERSTAND THE RIPPLES IN A FIR:

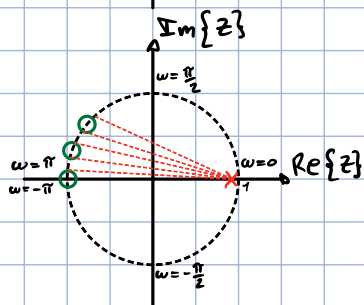
WHEREVER WE PLACE A ZERO ON THE Z-PLANE, THE TRANSFER FUNCTION WILL BE ZERO FOR THE CORRESPONDING FREQUENCIES.

CONSIDER THE FOLLOWING:



IN THIS CASE WE HAVE A ZERO IN $\omega=\pi$ AND WE HAVE THAT THE DISTANCE OF THE POLE FROM $r=1$ IS MONOTONICALLY INCREASING WITH ω ($\rightarrow$ LFF).

SUPPOSE NOW THAT WE WANT MORE "FLAT AREA" AROUND $\omega=\pi$, THE OBVIOUS THING WOULD BE TO ADD MORE ZEROS AROUND THERE.



NOW WE INDEED HAVE A FLATTER EXID, BUT WE INTRODUCED SOME RIPPLES. THIS IS BECAUSE IN THE GAPS BETWEEN ZEROS, THE RATIO OF DISTANCES IS $\neq 0$ EVEN THOUGH IT GOES TO ZERO AT EACH ZERO.

IT'S EASY NOW TO UNDERSTAND THAT MORE ZEROS LEAD TO A FLATTER RESPONSE BUT WITH INEVITABLE RIPPLES, UNLESS ∞ ZEROS.

WE CAN NOW WRITE THE AMPLITUDE RESPONSE:

$$|H(e^{j\omega})| = \left| \frac{b_0}{a_0} \right| \left| \frac{\prod_{k=1}^M Q_k}{\prod_{k=1}^N R_k} \right|$$

WHICH MEANS THAT THE AMPLITUDE RESPONSE DEPENDS ON THE RATIO OF THE PRODUCTS OF DISTANCES OF POLES AND ZEROS FROM A POINT ON THE UNIT CIRCLE DEFINED BY ω

SIMILARLY FOR THE PHASE:

$$\angle H(e^{j\omega}) = \left\lfloor \frac{b_0}{a_0} \right\rfloor - \omega(M-N) + \sum_{k=1}^M \theta_k - \sum_{k=1}^N \phi_k$$

LINEAR PHASE THAT DEPENDS ON DIFFERENCE BETWEEN M AND N
Ratio of Powers

SO

$$H(e^{j\omega}) = \underbrace{\left| \frac{b_0}{a_0} \right| \left| \frac{\prod_{k=1}^M Q_k}{\prod_{k=1}^N R_k} \right|}_{\text{AMPLITUDE RESPONSE}} e^{j \underbrace{\left\lfloor \frac{b_0}{a_0} \right\rfloor - \omega(M-N) + \sum_{k=1}^M \theta_k - \sum_{k=1}^N \phi_k}_{\text{PHASE RESPONSE}}}$$

SUPP. WE WANT A SPECIFIC FREQUENCY COMPLETELY SUPPRESSED. WE SIMPLY NEED TO PUT A ZERO ON THE UNIT CIRCLE AT THE DESIRED ω .

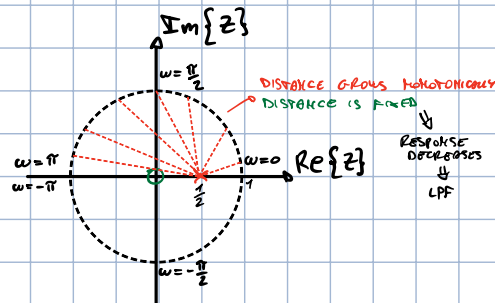
IF INSTEAD WE WANT TO ENHANCE A SPECIFIC FREQUENCY, WE JUST PUT A POLE CLOSE TO THE POINT ON THE UNIT CIRCLE (NOT ON THE UNIT CIRCLE, SINCE THE RESPONSE WOULD EXPLODE AND THE SYSTEM WOULDN'T BE STABLE ANYMORE)

EX

DRAW THE AMPLITUDE RESPONSE

$$H(z) = \frac{1}{1 - \frac{1}{2}z^{-1}} = \frac{z}{z - \frac{1}{2}} \rightarrow H(e^{j\omega}) = \frac{1}{1 - \frac{1}{2}e^{j\omega}}$$

POLES: $\frac{1}{2}$
ZEROS: $\emptyset$

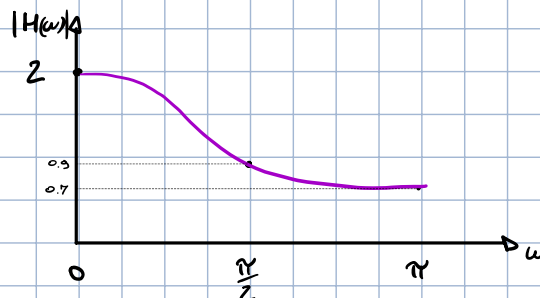


$$|H(e^{j\omega})| = \left| \frac{b_0}{a_0} \right| \left| \frac{\prod_{k=1}^M Q_k}{\prod_{k=1}^N R_k} \right|$$

$$|H(e^{j\omega})|_{\omega=0} = \frac{1}{1/2} = 2$$

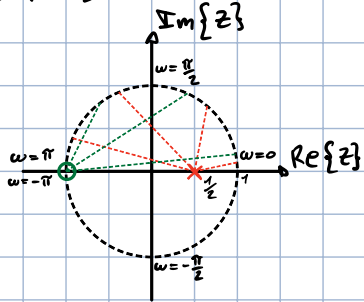
$$|H(e^{j\omega})|_{\omega=\pi/2} = \frac{1}{1 + (\frac{1}{2})^2} = \frac{2\sqrt{5}}{5} \approx 0.9$$

$$|H(e^{j\omega})|_{\omega=\pi} = \frac{1}{3/2} = \frac{2}{3} \approx 0.7$$



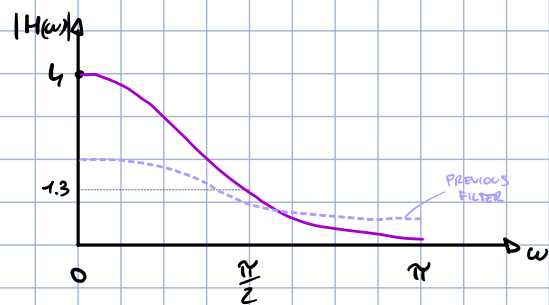
EX

DESIGN A FILTER THAT IS SHARPER THAN THE PREVIOUS ONE AND THAT HAS $\emptyset$ AMPLITUDE IN THE STOP-BAND



$$\omega=0: \frac{2}{1/2} = 4$$

$$\omega=\pi/2: \frac{\sqrt{2}}{\sqrt{1+(1/2)^2}} = \frac{2\sqrt{10}}{5} \approx 1.3$$



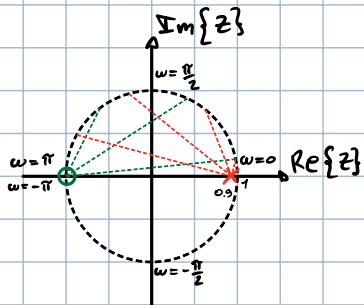
NOW ALSO THE DISTANCE OF THE ZERO DECREASES MONOTONICALLY, MAKING THE TRANSITION FASTER

WITH ZEROS AND POLES DEFINED, IT'S EASY TO FORMULATE THE Z-TRANSFORM (KEEP IN MIND THAT NOW WE ARE NOT RESPONDING IN ω BUT IN z)

POLES: $1/2$
ZEROS: -1

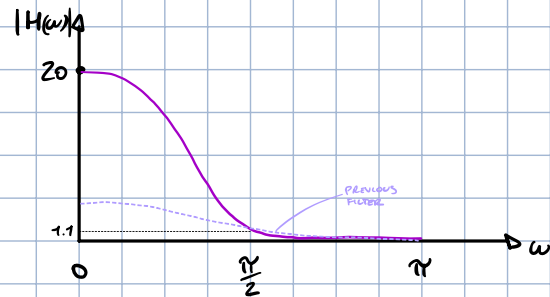
$$H(z) = \frac{z+1}{z-1/2}$$

TO MAKE THE TRANSITION EVEN FASTER WE CAN MOVE THE POLE CLOSER TO $r=1$



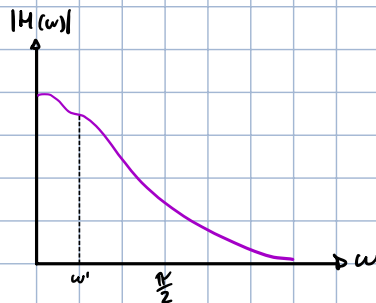
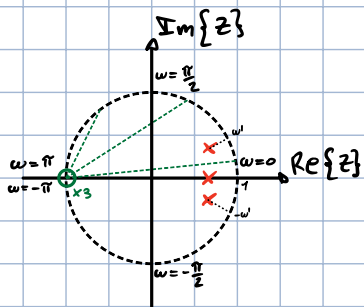
$$\omega=0: \frac{2}{0.1} = 20$$

$$\omega=\pi/2: \frac{\sqrt{2}}{\sqrt{1+0.81}} \approx 1.1$$



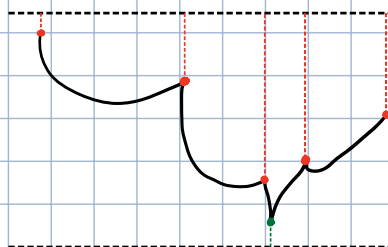
NOTE THAT THE AMPLITUDES IN $\omega=0$ ARE INCREASING AND THAT'S DUE TO THE POLE GETTING CLOSER AND CLOSER TO $\omega=0$

IF NOW WE WANT TO HAVE THE FILTER BROADER, WHILE KEEPING THE ZERO IN $\omega=\pi$, WE NEED TO ADD OTHER ZEROS/POLES



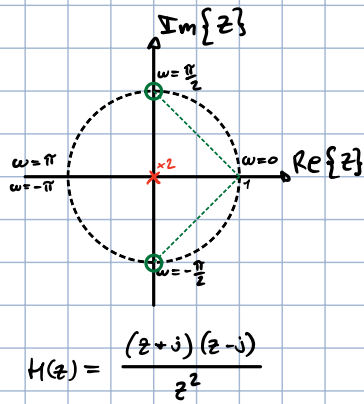
CAN THINK OF THE TRANSFER FUNCTION AS A CLOTH THAT IS HELD UP AT CERTAIN POINTS BY SOME SPRINGS. THESE ARE THE POLES.

THE ZEROS ARE SPRINGS THAT PULL IT DOWN INSTEAD.



EX

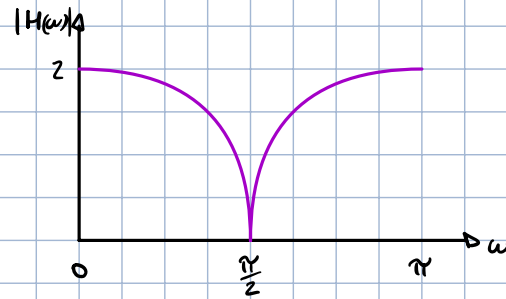
DESIGN A STOP-BAND FILTER THAT IS CAUSAL, STABLE AND REAL-VALUED



$$\omega=0: \frac{\sqrt{2}\sqrt{2}}{1 \cdot 1} = 2$$

$$\omega=\frac{\pi}{2}: \emptyset$$

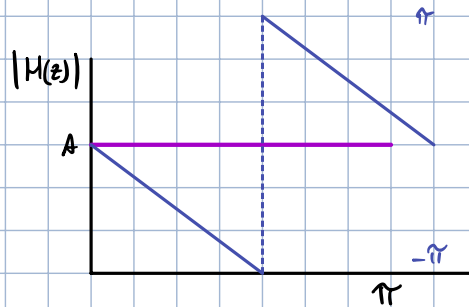
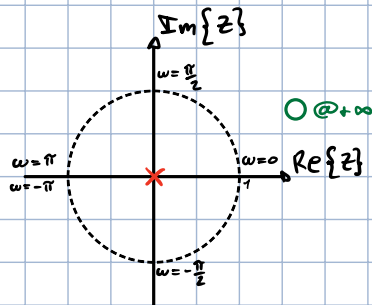
$$\omega=\pi: \frac{\sqrt{2}\sqrt{2}}{1 \cdot 1} = 2$$



EX

ALL-PASS FILTER

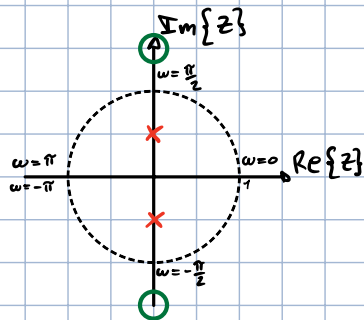
$$H(z) = A z^{-k} \rightarrow AS[n-k], \forall k$$



$$\begin{aligned} |H(\omega)| &= A \\ \angle H(\omega) &= -k\omega \end{aligned}$$

INVERTIBILITY AND MINIMUM-PHASE SYSTEMS

ALL-PASS IN A SICK WAY



POLES AND ZEROS ARE INVERSE COMPLEX CONJUGATE
 $\rightarrow$ THE DISTANCES ARE ALWAYS CONSTANT

$$H(z) = \frac{z^{-1} - p_k^*}{1 - p_k z^{-1}}$$

OBVIOUSLY THE POLES MUST STAY INSIDE UNIT CIRCLE, NOT OTHER WAY AROUND, OTHERWISE WOULD NOT BE STABLE.

THIS BECOMES A PROBLEM IF WE WANT ALSO THE INVERSE SYSTEM TO BE CAUSAL AND STABLE, SINCE IN THE INVERSE POLES BECOME ZEROS AND ZEROS BECOME POLES.

THERE IS A CLASS OF LTI SYSTEMS THAT IS STABLE AND CAUSAL AND ALSO HAS A STABLE AND CAUSAL INVERSE.

THIS HAPPENS IF AND ONLY IF BOTH POLES AND ZEROS ARE INSIDE THE UNIT CIRCLE. FILTERS OF THIS TYPE ARE CALLED "MINIMUM-PHASE SYSTEMS"

MINIMUM-PHASE AND ALL-PASS DECOMPOSITION

ANY CAUSAL AND STABLE SYSTEM (NO ZEROS ON UNIT CIRCLE) CAN BE DECOMPOSED INTO A MINIMUM-PHASE SYSTEM AND AN ALL-PASS SYSTEM

SUPP. THAT $H(z)$ HAS ONE ZERO IN $z = \frac{1}{a^*}$, $|a| < 1$ OUTSIDE THE UNIT CIRCLE AND ALL OTHER POLES/ZEROS ARE INSIDE THE UNIT CIRCLE.

$$H(z) = \frac{\prod_{n=0}^M (z - z_n)}{\prod_{n=0}^N (z - p_n)} = \frac{\prod_{n=0}^M (1 - z_n z^{-1})}{\prod_{n=0}^N (1 - p_n z^{-1})} = \underbrace{\frac{\prod_{n=0}^{M-1} (1 - z_n z^{-1})}{\prod_{n=0}^N (1 - p_n z^{-1})}}_{H_1(z)} (z^{-1} - \alpha^*) = H_1(z) (z^{-1} - \alpha^*)$$

← CANCEL POLE

NOTE THAT

$$H(z) \cdot \text{ALLPASS} = H(z) \frac{z^{-1} - \alpha^*}{1 - \alpha z^{-1}} = H(z)$$

THEN

$$H(z) = H_1(z) (z^{-1} - \alpha^*) \frac{1 - \alpha z^{-1}}{1 - \alpha z^{-1}} = H_1(z) (1 - \alpha z^{-1}) \frac{z^{-1} - \alpha^*}{1 - \alpha z^{-1}}$$

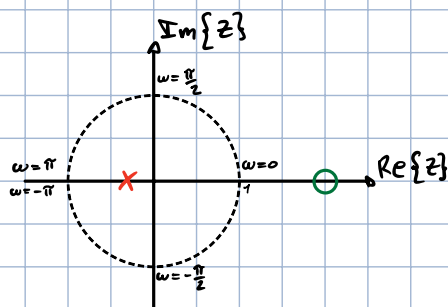
MINIMUM PHASE
↓
REMOVED CANCEL POLE
ALL-PASS

↑ MAY ALSO FIND IT AS THE INVERSE!

EX

$$H(z) = \frac{1 - 2z^{-1}}{1 + \frac{1}{3}z^{-1}} = \frac{z - 2}{z + \frac{1}{3}}$$

POLES: $-\frac{1}{3}$
ZEROS: 2



$$\frac{z-2}{z+\frac{1}{3}} = H_1(z) (\tilde{z}^{-1} - 2) \frac{1 - 2\tilde{z}^{-1}}{\tilde{z}^{-1} - 2} = \frac{1 - 2\tilde{z}^{-1}}{\tilde{z}^{-1} - 2} \cdot \frac{z-2}{1-2z}$$

MPF ALLPASS

$$H_{MP}(z) = \frac{1 - 2z}{z + \frac{1}{3}}$$

